Determining Factors for Delaying Medical Treatment

NHIS 2024 Multivariable Logistic Regression Analysis

Nandita Kannapadi

2026-07-19

```
library(dplyr)
library(ggplot2)
library(broom)
library(readr)
```

Research Question

Which factors — insurance coverage, race/ethnicity, or educational attainment — are most strongly associated with delaying medical care due to cost among U.S. adults? Using the 2024 National Health Interview Survey (NHIS) Sample Adult file, this analysis fits bivariate and multivariable logistic regression models to separate the independent contribution of each factor.

Reading the Data

```
# Data: NHIS 2024 Sample Adult file, downloaded from
# https://www.cdc.gov/nchs/nhis/documentation/2024-nhis.html
adult2024 <- read_csv("national_health_interview_survey.csv")

n_raw <- nrow(adult2024)
n_raw
```

```
## [1] 32629
```

Cleaning: Variable-Specific Missing Value Handling

`HISPALLP_A` and `EDUCP_A` use 97/98/99 for missing/non-response, while `MEDDL12M_A` and `NOTCOV_A` use 7/8/9. A single blanket `na.strings` won't work here because `EDUCP_A` also uses 8 and 9 as **valid** response categories (Bachelor's and Master's degree). So each variable is cleaned individually against its own codebook.

```
adult2024_clean <- adult2024 %>%
  mutate(
    HISPALLP_A = ifelse(HISPALLP_A %in% c(97, 98, 99), NA, HISPALLP_A),
    EDUCP_A    = ifelse(EDUCP_A %in% c(97, 98, 99), NA, EDUCP_A),
    MEDDL12M_A = ifelse(MEDDL12M_A %in% c(7, 8, 9), NA, MEDDL12M_A),
    NOTCOV_A   = ifelse(NOTCOV_A %in% c(7, 8, 9), NA, NOTCOV_A)
```

```r
  ) %>%
  filter(!is.na(HISPALLP_A), !is.na(EDUCP_A),
         !is.na(MEDDL12M_A), !is.na(NOTCOV_A))

n_clean <- nrow(adult2024_clean)
n_clean
```

```
## [1] 32142
```

After this step, the sample decreased from 32,629 to 32,142 observations.

Recoding and Restricting Categories

Race/ethnicity follows the NHIS `HISPALLP_A` codebook (1 = Hispanic, 2 = Non-Hispanic White, 3 = Non-Hispanic Black/African American, 4 = Non-Hispanic Asian); smaller multi-race and AIAN categories (codes 5–7) are excluded. Education is restricted to High School, Bachelor's, Master's, and Doctoral degree holders. Rows with missing data on any predictor of interest or the outcome are dropped (complete-case analysis).

```r
adult2024_recoded <- adult2024_clean %>%
  mutate(
    education = case_when(
      EDUCP_A == 4  ~ "High School Degree",
      EDUCP_A == 8  ~ "Bachelors Degree",
      EDUCP_A == 9  ~ "Masters Degree",
      EDUCP_A == 10 ~ "Doctoral Degree",
      TRUE ~ NA_character_
    ),
    insurance = case_when(
      NOTCOV_A == 1 ~ "Not covered",
      NOTCOV_A == 2 ~ "Covered",
      TRUE ~ NA_character_
    ),
    race_ethnicity = case_when(
      HISPALLP_A == 1 ~ "Hispanic",
      HISPALLP_A == 2 ~ "White",
      HISPALLP_A == 3 ~ "Black/African American",
      HISPALLP_A == 4 ~ "Asian",
      TRUE ~ NA_character_
    ),
    delayed_dummy = case_when(
      MEDDL12M_A == 1 ~ 1,
      MEDDL12M_A == 2 ~ 0,
      TRUE ~ NA_real_
    ),
    income_level = case_when(
      RATCAT_A %in% c(1, 2, 3)      ~ "Below poverty (<100%)",
      RATCAT_A %in% c(4, 5, 6, 7)   ~ "Near poverty (100-199%)",
      RATCAT_A %in% c(8, 9, 10, 11) ~ "Middle income (200-399%)",
      RATCAT_A %in% c(12, 13, 14)   ~ "Higher income (400%+)",
      TRUE ~ NA_character_
    ),
    income_level = factor(income_level,
```

```
      levels = c("Below poverty (<100%)", "Near poverty (100-199%)",
                 "Middle income (200-399%)", "Higher income (400%+)"))
  )

df_complete <- adult2024_recoded %>%
  filter(
    !is.na(education),
    !is.na(insurance),
    !is.na(race_ethnicity),
    !is.na(delayed_dummy)
  ) %>%
  mutate(
    education = factor(education,
      levels = c("High School Degree", "Bachelors Degree",
                 "Masters Degree", "Doctoral Degree")),
    insurance = factor(insurance, levels = c("Covered", "Not covered")),
    race_ethnicity = factor(race_ethnicity,
      levels = c("White", "Black/African American", "Asian", "Hispanic"))
  )

n_final <- nrow(df_complete)
n_final
```

```
## [1] 19303
```

The analytic sample contains 19,303 adults. The reduction from 32,142 reflects the deliberate category restrictions above (not solely item nonresponse): adults outside the four education and four race/ethnicity groups are excluded by design.

Descriptive Analysis

Table 1: Adults by Education

```
df_complete %>%
  count(education, name = "Count") %>%
  mutate(Percent = scales::percent(Count / sum(Count), accuracy = 0.1)) %>%
  rename(Education = education) %>%
  as.data.frame()
```

```
##            Education Count Percent
## 1 High School Degree  7301   37.8%
## 2   Bachelors Degree  7254   37.6%
## 3     Masters Degree  3528   18.3%
## 4    Doctoral Degree  1220    6.3%
```

Table 2: Adults by Race/Ethnicity

```r
df_complete %>%
  count(race_ethnicity, name = "Count") %>%
  mutate(Percent = scales::percent(Count / sum(Count), accuracy = 0.1)) %>%
  rename(`Race/Ethnicity` = race_ethnicity) %>%
  as.data.frame()
```

```
##           Race/Ethnicity Count Percent
## 1                  White 13726   71.1%
## 2 Black/African American  1716    8.9%
## 3                  Asian  1462    7.6%
## 4               Hispanic  2399   12.4%
```

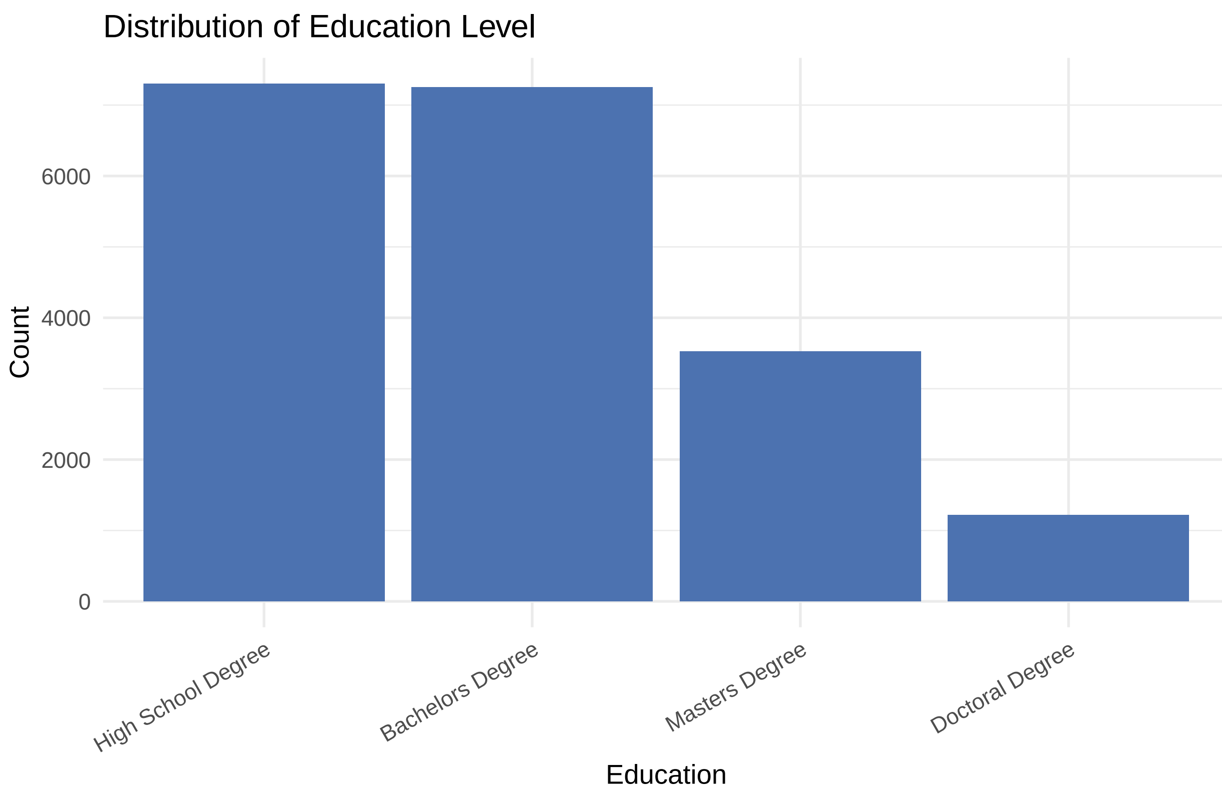
Table 3: Adults by Insurance Status

```r
df_complete %>%
  count(insurance, name = "Count") %>%
  mutate(Percent = scales::percent(Count / sum(Count), accuracy = 0.1)) %>%
  rename(`Insurance Status` = insurance) %>%
  as.data.frame()
```

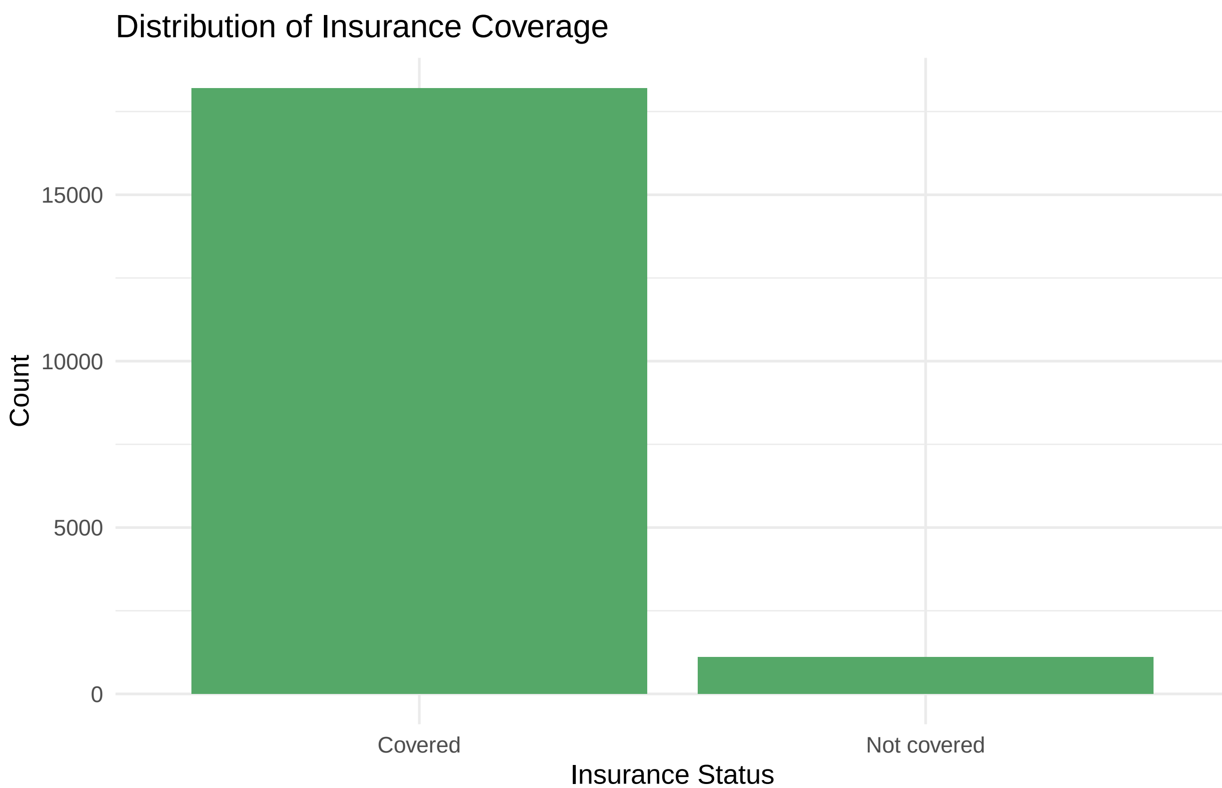
```
##   Insurance Status Count Percent
## 1          Covered 18203   94.3%
## 2      Not covered  1100    5.7%
```

Distribution Plots

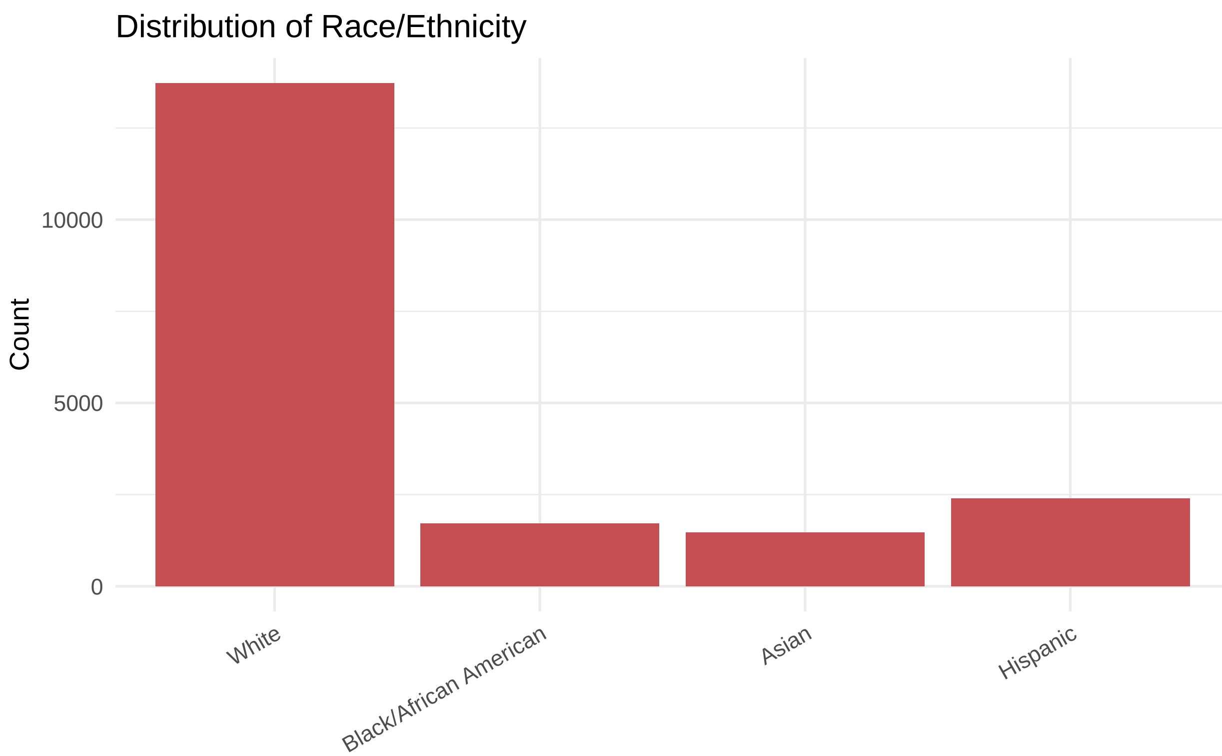
```r
ggplot(df_complete, aes(x = education)) +
  geom_bar(fill = "#4C72B0") +
  labs(title = "Distribution of Education Level",
       x = "Education", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



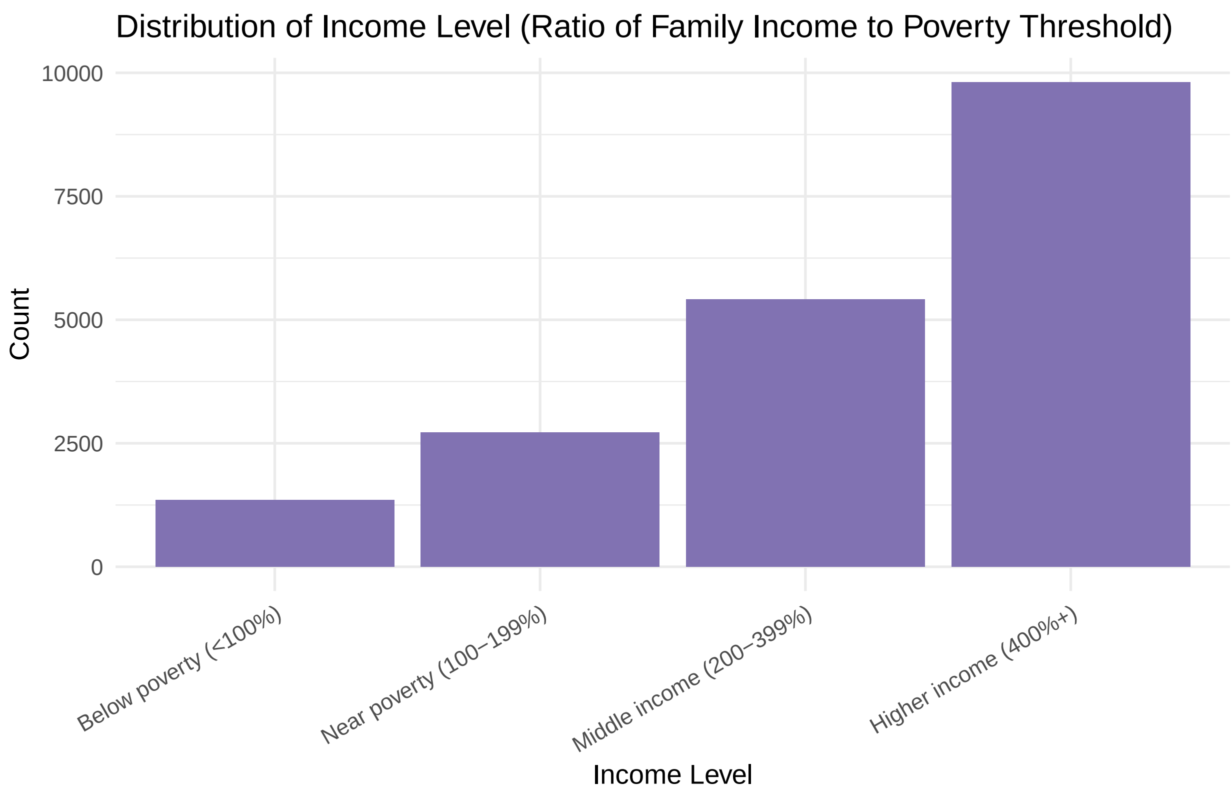

```r
ggplot(df_complete, aes(x = insurance)) +
  geom_bar(fill = "#55A868") +
  labs(title = "Distribution of Insurance Coverage",
       x = "Insurance Status", y = "Count") +
  theme_minimal()
```




```r
ggplot(df_complete, aes(x = race_ethnicity)) +
  geom_bar(fill = "#C44E52") +
  labs(title = "Distribution of Race/Ethnicity",
       x = "Race/Ethnicity", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```




```
ggplot(df_complete, aes(x = income_level)) +
  geom_bar(fill = "#8172B2") +
  labs(title = "Distribution of Income Level (Ratio of Family Income to Poverty Threshold)",
       x = "Income Level", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



Income is summarized here for descriptive context; it is not included in the regression models below, which focus on the three hypothesized predictors. Adding income as a covariate is a natural extension.

Proportions of Delayed Care by Group

Table 4: Delayed Care by Insurance Status

```
df_complete %>%
  group_by(insurance) %>%
  summarize(n = n(), prop_delayed = mean(delayed_dummy)) %>%
  mutate(prop_delayed = scales::percent(prop_delayed, accuracy = 0.1)) %>%
  rename(`Insurance Status` = insurance, N = n,
         `% Delayed Care` = prop_delayed) %>%
  as.data.frame()
```

```
##   Insurance Status     N % Delayed Care
## 1          Covered 18203           5.8%
## 2      Not covered  1100          27.5%
```

Table 5: Delayed Care by Education

```
df_complete %>%
  group_by(education) %>%
  summarize(n = n(), prop_delayed = mean(delayed_dummy)) %>%
```

```r
  mutate(prop_delayed = scales::percent(prop_delayed, accuracy = 0.1)) %>%
  rename(Education = education, N = n,
         `% Delayed Care` = prop_delayed) %>%
  as.data.frame()
```

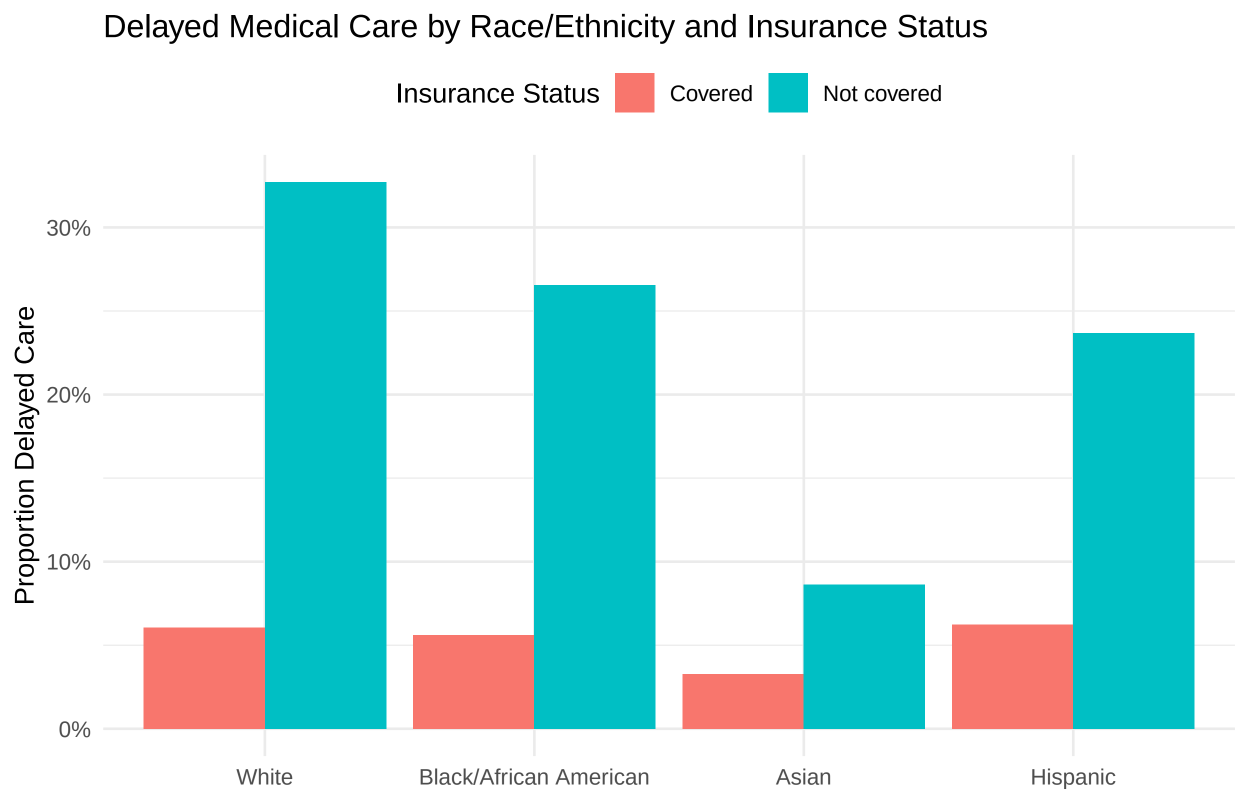
```
##            Education    N % Delayed Care
## 1 High School Degree 7301           7.5%
## 2   Bachelors Degree 7254           7.1%
## 3     Masters Degree 3528           7.1%
## 4    Doctoral Degree 1220           4.3%
```

Table 6: Delayed Care by Race/Ethnicity and Insurance Status

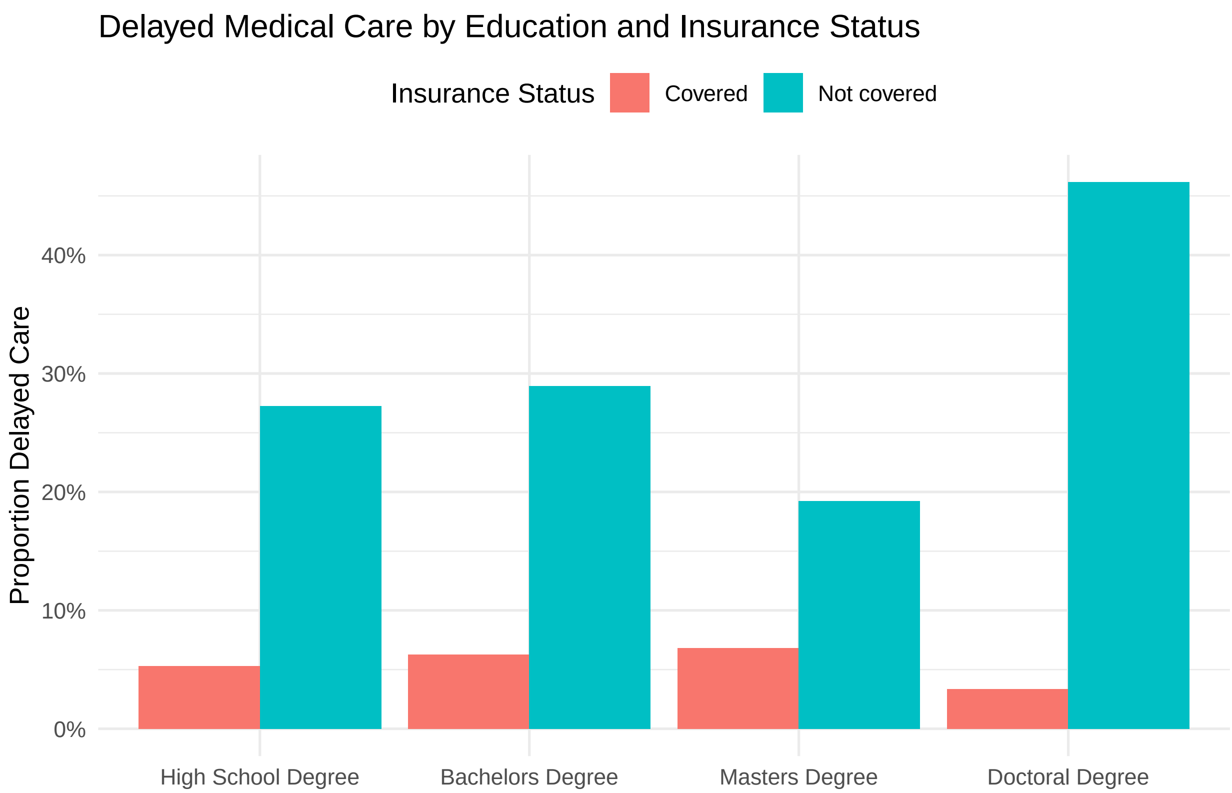
```r
df_complete %>%
  group_by(race_ethnicity, insurance) %>%
  summarize(n = n(),
            prop_delayed = mean(delayed_dummy),
            .groups = "drop") %>%
  mutate(prop_delayed = scales::percent(prop_delayed, accuracy = 0.1)) %>%
  rename(`Race/Ethnicity` = race_ethnicity,
         `Insurance Status` = insurance, N = n,
         `% Delayed Care` = prop_delayed) %>%
  as.data.frame()
```

```
##           Race/Ethnicity Insurance Status     N % Delayed Care
## 1                  White          Covered 13194           6.1%
## 2                  White      Not covered   532          32.7%
## 3 Black/African American          Covered  1603           5.6%
## 4 Black/African American      Not covered   113          26.5%
## 5                  Asian          Covered  1404           3.3%
## 6                  Asian      Not covered    58           8.6%
## 7               Hispanic          Covered  2002           6.2%
## 8               Hispanic      Not covered   397          23.7%
```

```r
df_complete %>%
  group_by(race_ethnicity, insurance) %>%
  summarize(prop_delayed = mean(delayed_dummy), .groups = "drop") %>%
  ggplot(aes(x = race_ethnicity, y = prop_delayed, fill = insurance)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Delayed Medical Care by Race/Ethnicity and Insurance Status",
    x = NULL,
    y = "Proportion Delayed Care",
    fill = "Insurance Status"
  ) +
  theme_minimal() +
  theme(legend.position = "top")
```




```r
df_complete %>%
  group_by(education, insurance) %>%
  summarize(prop_delayed = mean(delayed_dummy), .groups = "drop") %>%
  ggplot(aes(x = education, y = prop_delayed, fill = insurance)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Delayed Medical Care by Education and Insurance Status",
    x = NULL,
    y = "Proportion Delayed Care",
    fill = "Insurance Status"
  ) +
  theme_minimal() +
  theme(legend.position = "top")
```



Descriptive takeaways. Differences in delayed care across race/ethnicity groups are modest among insured adults but widen considerably among the uninsured, suggesting insurance coverage — not race/ethnicity itself — may drive much of the raw disparity. Cell sizes matter when reading the uninsured bars: several race-by-insurance cells contain fewer than 150 respondents (see Table 6), so those proportions carry wide uncertainty and should be interpreted cautiously. The formal models below test these patterns directly.

Likelihood Ratio Tests: Bivariate Models vs. Null Model

Each predictor is first tested on its own against an intercept-only model.

```
null_model <- glm(delayed_dummy ~ 1, family = "binomial",
                  data = df_complete)

model_education_biv <- glm(delayed_dummy ~ education,
                           family = "binomial", data = df_complete)
model_race_biv      <- glm(delayed_dummy ~ race_ethnicity,
                           family = "binomial", data = df_complete)
model_insurance_biv <- glm(delayed_dummy ~ insurance,
                           family = "binomial", data = df_complete)

lr_row <- function(model, label) {
  a <- anova(null_model, model, test = "Chisq")
  tibble(
    Predictor = label,
    `Chi-sq`  = a$Deviance[2],
    df        = a$Df[2],
    `p-value` = format.pval(a$`Pr(>Chi)`[2], digits = 3, eps = 0.001)
```

```r
  )
}

lrt_table <- bind_rows(
  lr_row(model_education_biv, "Education"),
  lr_row(model_race_biv, "Race/Ethnicity"),
  lr_row(model_insurance_biv, "Insurance")
)

# Likelihood ratio tests of each bivariate model against the null model
lrt_table %>%
  mutate(`Chi-sq` = round(`Chi-sq`, 2)) %>%
  as.data.frame()
```

```
##        Predictor Chi-sq df p-value
## 1      Education  18.66  3  <0.001
## 2 Race/Ethnicity  48.97  3  <0.001
## 3      Insurance 472.23  1  <0.001
```

Multivariable Logistic Regression

```r
model_multivariate <- glm(
  delayed_dummy ~ insurance + race_ethnicity + education,
  family = "binomial",
  data = df_complete
)

lr_full <- lr_stat(model_multivariate)
anova(null_model, model_multivariate, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: delayed_dummy ~ 1
## Model 2: delayed_dummy ~ insurance + race_ethnicity + education
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1     19302     9863.2
## 2     19295     9335.4  7   527.73 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
term_labels <- c(
  "insuranceNot covered" = "Uninsured (vs. Covered)",
  "race_ethnicityBlack/African American" =
    "Black/African American (vs. White)",
  "race_ethnicityAsian" = "Asian (vs. White)",
  "race_ethnicityHispanic" = "Hispanic (vs. White)",
  "educationBachelors Degree" = "Bachelor's (vs. High School)",
  "educationMasters Degree" = "Master's (vs. High School)",
  "educationDoctoral Degree" = "Doctoral (vs. High School)"
)
```

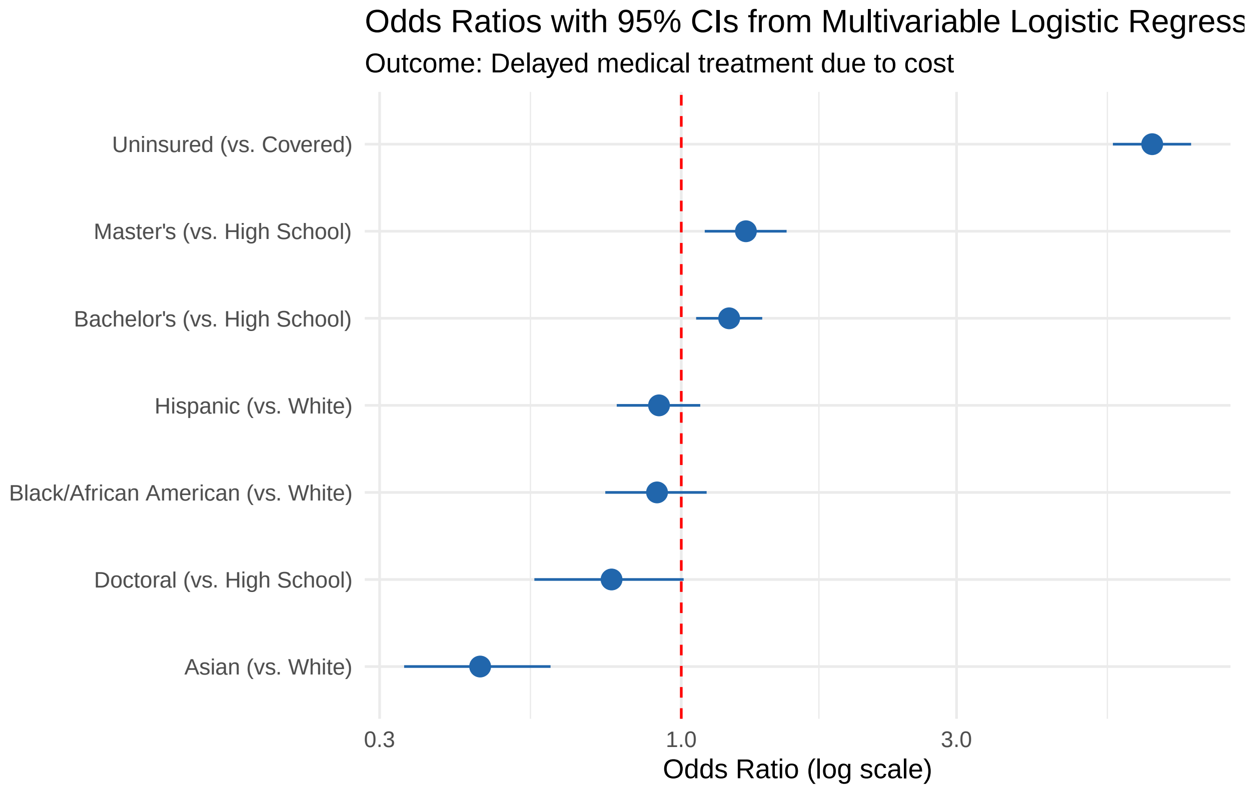
```r
or_table <- tidy(model_multivariate, exponentiate = TRUE,
                 conf.int = TRUE) %>%
  filter(term != "(Intercept)") %>%
  mutate(Predictor = recode(term, !!!term_labels))

or_table %>%
  transmute(
    Predictor,
    `Odds Ratio` = sprintf("%.2f", estimate),
    `95% CI` = sprintf("[%.2f, %.2f]", conf.low, conf.high),
    `p-value` = format.pval(p.value, digits = 3, eps = 0.001)
  ) %>%
  as.data.frame()
```

```
##                            Predictor Odds Ratio       95% CI p-value
## 1            Uninsured (vs. Covered)       6.55 [5.60, 7.65] < 0.001
## 2 Black/African American (vs. White)       0.91 [0.74, 1.11] 0.34790
## 3                      Asian (vs. White)       0.45 [0.33, 0.59] < 0.001
## 4                   Hispanic (vs. White)       0.91 [0.77, 1.08] 0.29588
## 5        Bachelor's (vs. High School)       1.21 [1.06, 1.38] 0.00452
## 6          Master's (vs. High School)       1.29 [1.10, 1.52] 0.00195
## 7          Doctoral (vs. High School)       0.76 [0.56, 1.01] 0.06659
```

Odds Ratio Forest Plot

```r
ggplot(or_table,
       aes(x = reorder(Predictor, estimate), y = estimate)) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high),
                  color = "#2166AC", size = 0.7) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  scale_y_log10() +
  coord_flip() +
  labs(
    title = "Odds Ratios with 95% CIs from Multivariable Logistic Regression",
    subtitle = "Outcome: Delayed medical treatment due to cost",
    x = NULL,
    y = "Odds Ratio (log scale)"
  ) +
  theme_minimal()
```



Interpretation and Discussion

Bivariate models. All three predictors significantly improved model fit over the null model. Education showed the weakest association ($\chi^2 = 18.66$, $p < .001$), race/ethnicity was stronger ($\chi^2 = 48.97$, $p < .001$), and insurance coverage showed the strongest association by far ($\chi^2 = 472.23$, $p < .001$).

Multivariable model. The full model with all three predictors significantly improved fit over the null model ($\chi^2 = 527.73$, $p < .001$).

- **Insurance** was the strongest predictor: uninsured adults had about 6.55 times the odds of delaying care compared to insured adults (95% CI [5.60, 7.65], $p < .001$), even after controlling for race/ethnicity and education.
- **Race/ethnicity**: after adjustment, Black/African American adults (OR = 0.91, p = 0.348) and Hispanic adults (OR = 0.91, p = 0.296) did not differ significantly from White adults. Asian adults were the exception, with about 55% lower odds than White adults (OR = 0.45, $p < .001$).
- **Education**: contrary to the original hypothesis, Bachelor's degree holders had higher odds of delaying care than high school graduates (OR = 1.21, p = 0.005), as did Master's degree holders (OR = 1.29, p = 0.002). Doctoral degree holders trended lower (OR = 0.76, p = 0.067).

Takeaway. Raw racial disparities in delayed care appear to be driven largely by unequal insurance access rather than race/ethnicity itself: group differences attenuate substantially once insurance and education are controlled for. Insurance coverage remains the most direct policy lever for reducing cost-related care delays.

Limitations

- **Unweighted estimates.** This analysis does not apply NHIS survey weights (`WTFA_A`) or account for the complex sampling design, so estimates describe the analytic sample rather than the U.S. adult population. A design-based extension using the `survey` package would produce population-representative estimates.

- **Complete-case analysis.** Listwise deletion and deliberate category restrictions reduce the sample and could introduce selection effects if missingness is not random.
- **Cross-sectional data.** Associations are not causal; unmeasured confounders such as income, health status, and geography likely influence both coverage and care-seeking. Income in particular is a strong candidate covariate for future model iterations.